Experiment 6: Tracking ML Experiments using MLflow with Azure ML Integration

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AIM

To study experiment tracking using MLflow and its integration with Azure Machine Learning.

OBJECTIVES

1. To understand the concept of experiment tracking in ML projects.
2. To study offline setup using MLflow for local experiment tracking.
3. To understand integration of MLflow with Azure ML for cloud-based tracking.

THEORY

1. Experiment Tracking in Machine Learning

In ML projects, multiple experiments are performed by changing parameters, models, or datasets. Tracking these experiments is important to compare performance and reproduce results.

MLflow is an open-source platform used to log parameters, metrics, models, and artifacts. It helps maintain organized records of all experiments for better analysis and reproducibility.

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2. Offline Setup: MLflow Local Tracking

In the offline setup, MLflow runs locally and provides a web-based UI to monitor experiments. Developers can log metrics such as accuracy and loss, compare runs, and store trained models.

The MLflow UI on localhost allows visualization and management of experiments during development or in environments like Google Colab. This ensures systematic tracking before moving to production.

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3. Azure ML and MLflow Integration

Microsoft Azure provides

Azure Machine Learning, which supports native integration with MLflow.

Azure ML allows MLflow tracking through a remote tracking URI. This enables experiments to be logged directly into Azure ML Workspace, supporting collaboration, centralized tracking, and scalable storage.

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4. End-to-End Workflow

In practice, experiments are first tested and tracked locally using MLflow. After validation, MLflow is connected to Azure ML Workspace to log experiments in the cloud.

This hybrid approach ensures reproducibility, centralized experiment management, and seamless transition from local development to scalable cloud tracking.

CONCLUSION

In this experiment, we studied experiment tracking using MLflow and its integration with Azure Machine Learning. We understood local tracking through MLflow UI and cloud-based tracking using Azure ML integration.

This approach improves organization, reproducibility, and collaboration in Machine Learning workflows.